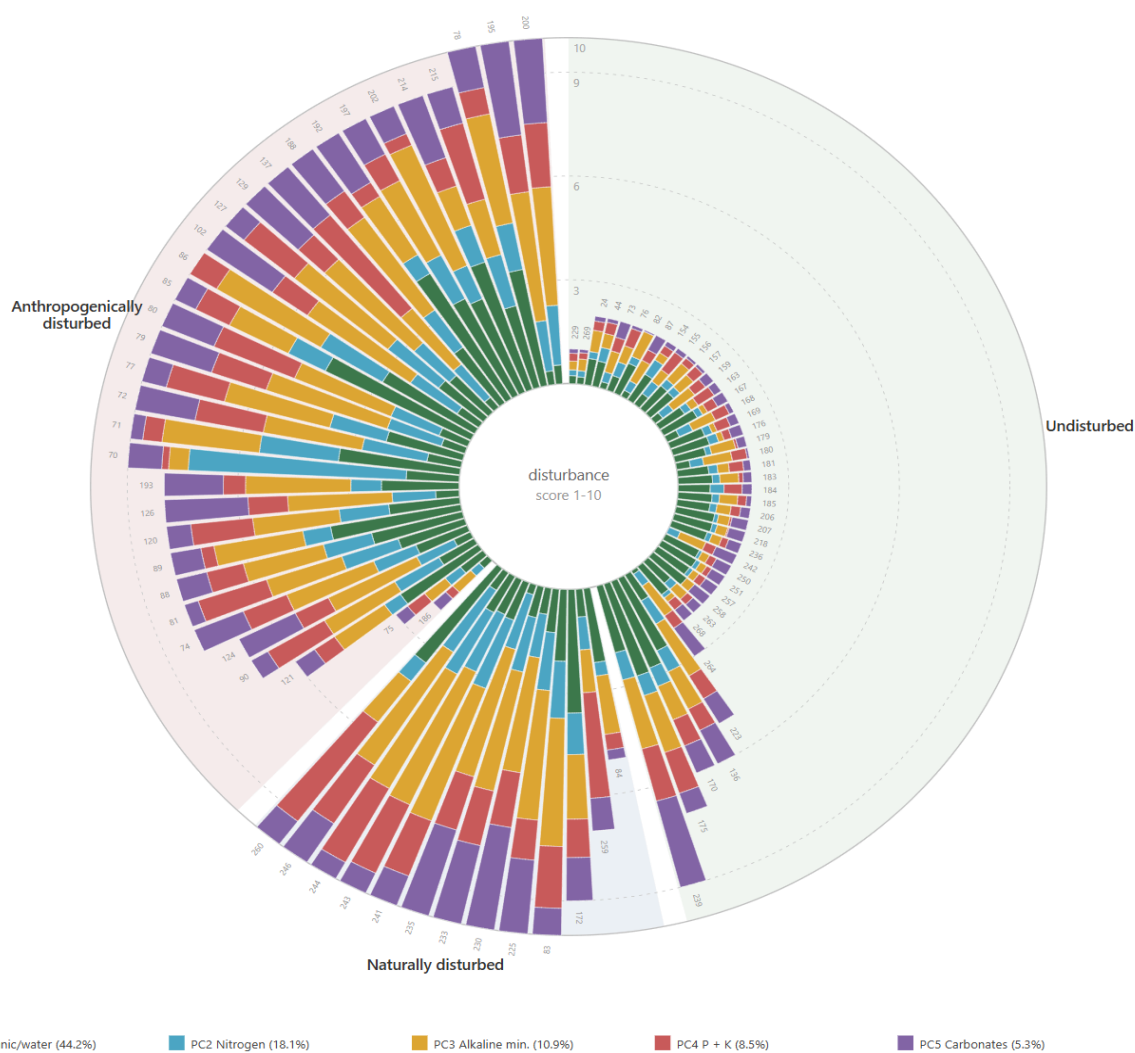


# Data Visualisation Project Report - Communication

## Part 1. Metadata

• Athulya Mohandas Menon (AMM, r1070436)

## Part 2. Original Visual

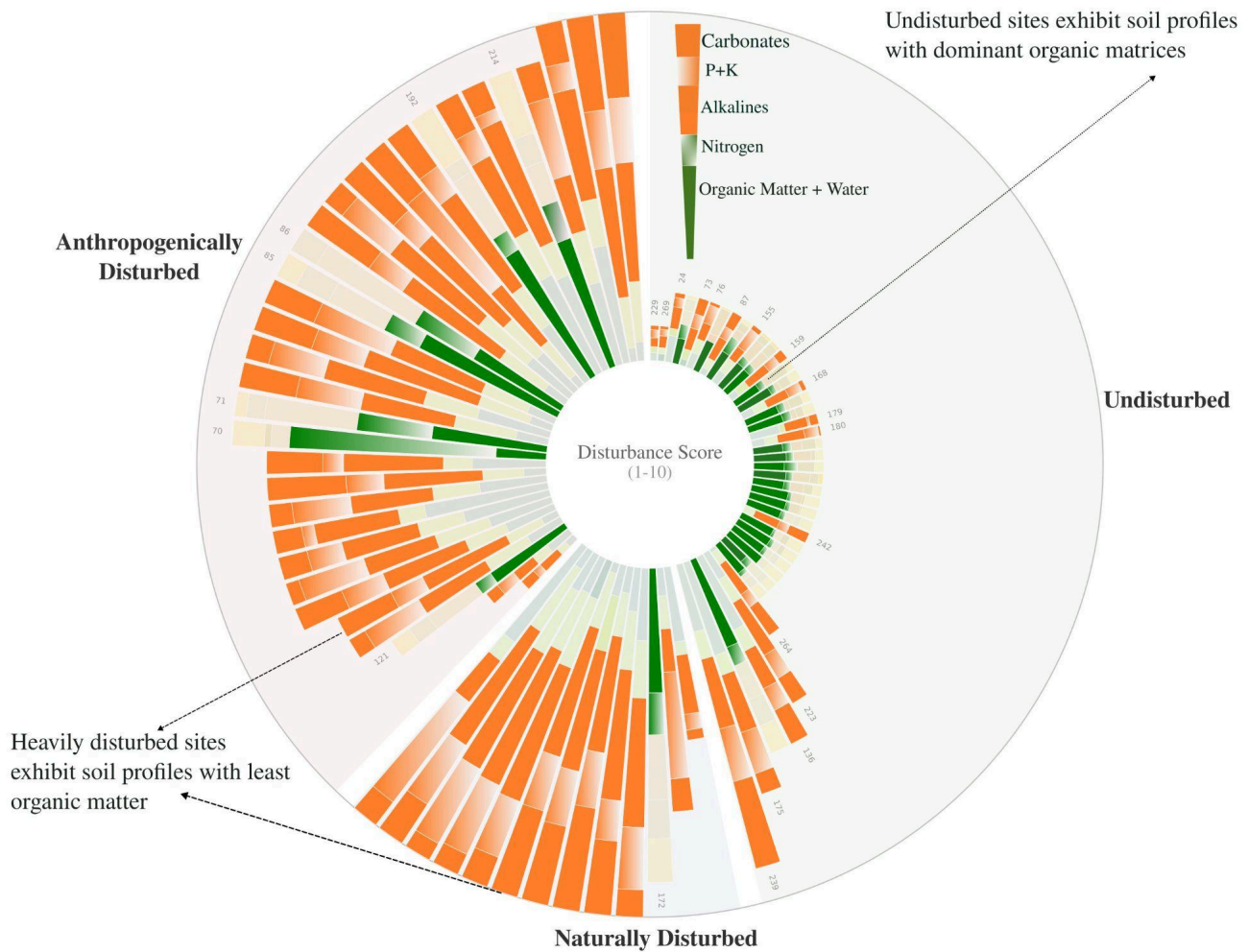


## Part 3. Final Design

**Title: Disturbance Degrades Arctic Soil Diversity - Comparative Radial Profiles**

**Caption:** Radial profile comparison of the observed 87 Oumalik plots sorted by ecological status(disturbance type and score) with natural and anthropogenically disturbed plots on the left hemisphere and undisturbed ones on the right hemisphere where each wedge displays the soil profile of the plot.

## Disturbance Degrades Arctic Soil Diversity - Comparative Radial Profiles



### Part 4. Description of the message

The visual is designed to convey the message that Arctic soil profile divergence and hence, the road to recovery is driven by organic matter accumulation, which supports biodiversity. It can be seen through the plot that undisturbed sites with lower disturbance scores exhibit higher organic matter and water-retention capacities. In contrast, both natural and human-induced disturbances (which also lead to higher disturbance scores) strip the soil of organic matter, leaving them with mineral deposits that require more effort for restoration. Instead of treating soil composition as a large-number of contrasting independent variables, this visual groups them by functional domains which helps in revealing an asymmetry. While macro-nutrients such as alkaline, potassium and phosphorus remain constant, and non-differentiating across sampled plots, the landscape splits into two based on organic accumulation. By highlighting this, the chart drives home the point that physical organic buildup should be the primary focus during restoration since it contributes the most to soil heterogeneity and habitat divergence in the tundra ecosystem.

## Part 5. Description of the target audience

The target audience consists of restoration ecologists and field biologists concerned with tundra conservation that are either attending a scientific session or reading an ecological journal. While such an audience possesses a strong understanding of soil chemistry, laying out all 87 plots with all independent ecological variables might be over-stimulating for them and would not help them understand the ecological and topological cost of disturbances in this specific area. The figure simplifies the visual, allowing the specialists to instantly comprehend the impact of different disturbances on the organic-to-chemical soil composition ratio of the Oumalik plots.

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## Part 6. Description of the design choices and rationale

### *Decluttering and Grounding - Subtraction*

The primary goal of the redesign was to eliminate the non-data ink (unimportant visual elements) used in the exploratory visual, leaving behind an easily scannable layout. The original concentric grid lines were removed to reduce visual noise and make the magnitudes of the stacked bars clearer. The different colours which were all high contrast were also removed.

### *Strategic Grouping*

The variables were then organized radially outward into two functional, high-contrast domains with the outer segment capturing the mineral pool, and the inner the organic matrix. Applying the principle of visual emphasis, the less important stack for each plot was transformed into a highly desaturated, low-opacity tone, leveraging color saturation and contrast to instantly guide the viewer's eye to the more vital profile for each plot, which creates a figure-ground relationship and value contrast.

### *Typographic Cleanup - Subtraction*

All the labels were made to follow a hierarchy and were labeled such that they can be read left-to-right with minimal head-movement to boost readability. Since the visual is to track trends, having all 87 plot numbers on the outside of the wheel felt redundant and hence to eliminate visual clutter, only the ones that seemed anomalous to their disturbance zone was retained so the ecologists may zoom in on them later to identify why they were different to the trend.

### *Integrated Key*

The empty right hemisphere was used to house a detailed breakdown of a single wedge acting as the key, helping to remove unnecessary legends and label the data directly, explaining marks and channels natively. This enables users to read the visual without having to constantly move their eyes to the legend to reassess labelling. The middle section was given a linear shading to help differentiate between the mineral and organic components while retaining visual cohesion and simplicity.

### *Semicircle Shading*

The background of the circular canvas post split was tinted to isolate differences between the disturbed and undisturbed plots using warm and green tones which helps capitalise on the viewer's pre-attentive focus (spatial grouping and background tone) to divide the visual.

### *Annotation Layer*

Have a clear chart title and further annotations on the empty white-space to justify the message and transform data into an active argument. The pointer lines track the annotations to the tips of the organic peaks helping the viewer see clearly that organic matrices should be the focal point of restoration efforts as per Amanda Cox's principles.